

Dynamic Webpage Hosting Using PhpMyAdmin

1. Objective

To install and configure **Apache HTTP Server, PHP, and MariaDB** on a Red Hat Enterprise Linux server and host a PHP-based website. The setup is then extended with **phpMyAdmin** for web-based management of the MariaDB database.

2. Environment

Component	Details
Operating System	Red Hat Enterprise Linux 10.2
Architecture	x86_64
Web Server	Apache HTTP Server
Database Server	MariaDB
PHP Version	8.3.33
phpMyAdmin Version	5.2.3
EC2 Public IP	13.201.184.251
Web Root	/var/www/html
Database	sampledb
Database User	webuser

3. Installing Apache

Apache HTTP Server was installed using the DNF package manager.

```
sudo dnf install httpd -y
```

The Apache service was started and enabled:

```
sudo systemctl start httpd
sudo systemctl enable httpd
```

Apache Configuration Files

Main configuration file:

`/etc/httpd/conf/httpd.conf`

Additional configuration directory:

`/etc/httpd/conf.d/`

The default Apache DocumentRoot is:

`/var/www/html`

4. Installing PHP

PHP and the required modules for MariaDB connectivity were installed:

```
sudo dnf install php php-cli php-fpm php-mysqlnd -y
```

The installed PHP version was:

PHP 8.3.33

PHP Configuration

Main PHP configuration file:

`/etc/php.ini`

Additional PHP module configuration:

`/etc/php.d/`

Important extensions include:

- `mysqli`
- `mysqlnd`
- `PDO`
- `pdo_mysql`
- `mbstring`
- `curl`
- `xml`

- opcache

5. Hosting the PHP Website

A PHP information page was created in the Apache DocumentRoot.

File:

`/var/www/html/index.php`

The following PHP code was used:

```
<?php
phpinfo();
?>
```

Apache was restarted:

```
sudo systemctl restart httpd
```

The PHP page was tested locally:

```
curl http://localhost/index.php
```

The website was then accessed through the EC2 public IP:

```
http://13.201.184.251/index.php
```

Output



PHP Version 8.3.33	
System	Linux ip-172-31-0-203.ap-south-1.compute.internal 6.12.0-211.16.1.el10_2.x86_64 #1 SMP PREEMPT_DYNAMIC Mon May 18 10:23:57 EDT 2026 x86_64
Build Date	Jul 28 2026 17:56:10
Build System	Red Hat Enterprise Linux release 10.2 (Coughlan)
Build Provider	Red Hat, Inc.
Compiler	gcc (GCC) 14.3.1 20251022 (Red Hat 14.3.1-4)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-ldap.ini, /etc/php.d/20-ldap_sasl.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-mysqlnd.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sqlite3.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-xsl.ini, /etc/php.d/30-mysqli.ini, /etc/php.d/30-pdo_mysql.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmldb.ini, /etc/php.d/30-xmldb_reader.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831.NTS
PHP Extension Build	API20230831.NTS

Figure 1: PHP information page successfully hosted through Apache

6. Installing MariaDB

MariaDB Server was installed:

```
sudo dnf install mariadb-server -y
```

The MariaDB service was started:

```
sudo systemctl start mariadb
```

The service was enabled:

```
sudo systemctl enable mariadb
```

The MariaDB client version was:

15.1

MariaDB Configuration

Main configuration file:

/etc/my.cnf

7. Securing MariaDB

The MariaDB security configuration was performed using:

```
sudo mariadb-secure-installation
```

The following security settings were applied:

- Unix socket authentication enabled for the root account
- Anonymous users removed
- Remote root login disabled
- Test database removed
- Privilege tables reloaded

8. Creating the Database and User

MariaDB was accessed using:

```
sudo mariadb
```

The database was created:

```
CREATE DATABASE sampled;
```

A dedicated database user was created:

```
CREATE USER 'webuser'@'localhost' IDENTIFIED BY '123';
```

Privileges were granted:

```
GRANT ALL PRIVILEGES ON sampled.* TO 'webuser'@'localhost';
```

Privileges were reloaded:

```
FLUSH PRIVILEGES;
```

9. Creating the Users Table

The database was selected:

```
USE sampledb;
```

The users table was created:

```
CREATE TABLE users (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100),  
    email VARCHAR(150)  
);
```

Sample records were inserted:

```
INSERT INTO users (name, email) VALUES  
( 'Sanjid', 'sanjid@example.com'),  
( 'Alex', 'alex@example.com');
```

The resulting database structure was:

```
sampledb  
├─ users  
│   ├── id  
│   ├── name  
│   └─ email
```

10. Installing phpMyAdmin

phpMyAdmin was installed to provide a web-based interface for managing MariaDB.

Since a suitable phpMyAdmin package was not available through the configured DNF repositories, the official phpMyAdmin archive was downloaded and extracted.

The phpMyAdmin files were placed in:

```
/var/www/html/phpmyadmin
```

A configuration file was created:

```
sudo cp /var/www/html/phpmyadmin/config.sample.inc.php  
/var/www/html/phpmyadmin/config.inc.php
```

A Blowfish secret was generated and configured in:

```
/var/www/html/phpmyadmin/config.inc.php
```

The phpMyAdmin directory was assigned to Apache:

```
sudo chown -R apache:apache /var/www/html/phpmyadmin
```

11. SELinux Configuration for phpMyAdmin

Initially, phpMyAdmin returned:

```
HTTP/1.1 403 Forbidden
```

The phpMyAdmin files had an incorrect SELinux context because they were initially extracted under /tmp.

The incorrect context was:

```
user_tmp_t
```

The appropriate context for Apache web content is:

```
httpd_sys_content_t
```

A persistent SELinux file context was configured:

```
sudo semanage fcontext -a -t httpd_sys_content_t  
"/var/www/html/phpmyadmin(/.*)?"
```

The context was applied:

```
sudo restorecon -Rv /var/www/html/phpmyadmin
```

After applying the correct SELinux context, phpMyAdmin became accessible.

12. Verifying phpMyAdmin

phpMyAdmin was tested locally:

```
curl -I http://localhost/phpmyadmin/
```

The response returned:

```
HTTP/1.1 200 OK
```

This confirmed that Apache was successfully serving phpMyAdmin and PHP was processing it.

phpMyAdmin was accessed through:

<http://13.201.184.251/phpmyadmin/>

The login credentials were:

Username: webuser

Password: 123

13. phpMyAdmin Database Verification

After logging into phpMyAdmin, the `samp1edb` database was opened.

The `users` table was displayed successfully with the sample records.

ID	Name	Email
1	Sanjid	sanjid@example.com
2	Alex	alex@example.com

Output

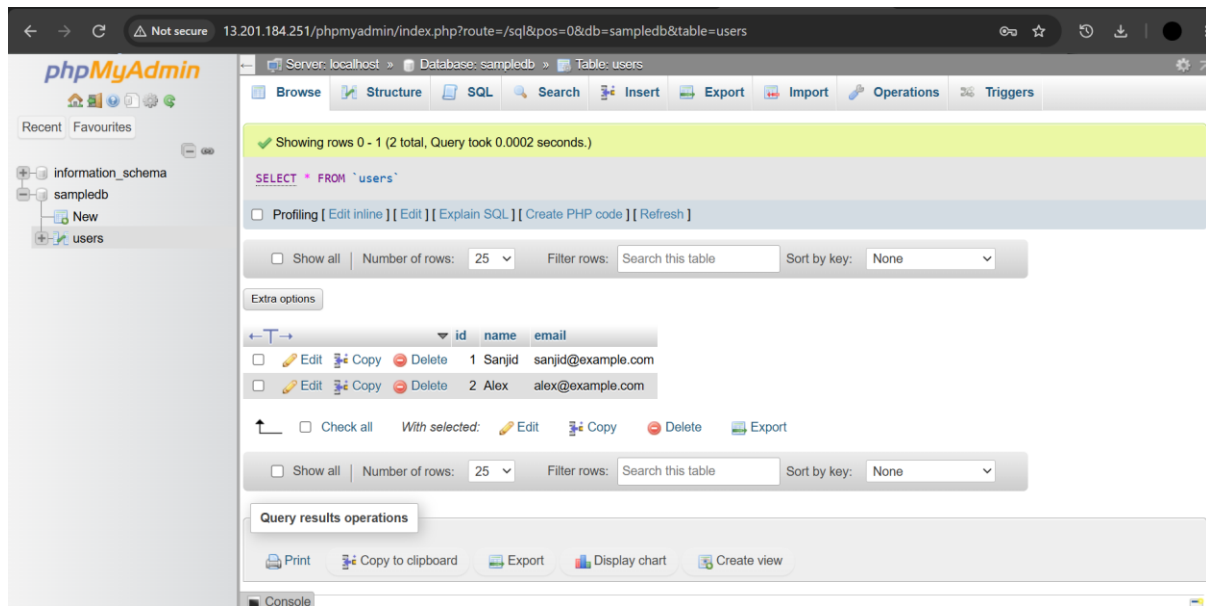


Figure 2: users table and sample records displayed in phpMyAdmin

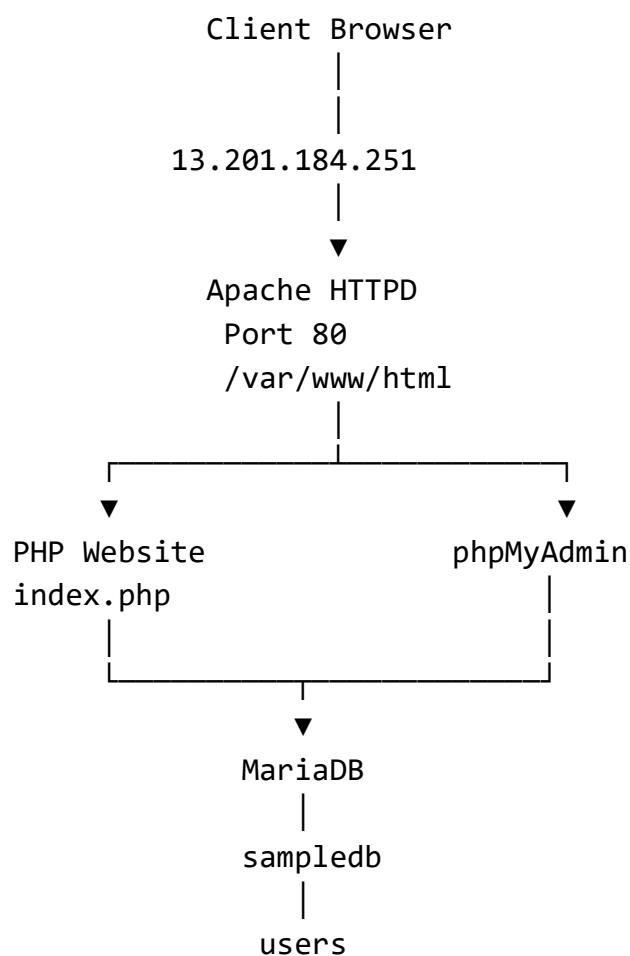
14. Important Configuration Files

Component	Configuration / Location
Apache main configuration	/etc/httpd/conf/httpd.conf
Apache additional configuration	/etc/httpd/conf.d/
Apache DocumentRoot	/var/www/html
PHP configuration	/etc/php.ini
PHP modules	/etc/php.d/
MariaDB configuration	/etc/my.cnf
phpMyAdmin configuration	/var/www/html/phpmyadmin/config.inc.php
PHP website	/var/www/html/index.php
phpMyAdmin directory	/var/www/html/phpmyadmin

15. Software Versions

Apache HTTP Server : 2.4.63
PHP : 8.3.33
MariaDB Client : 15.1
phpMyAdmin : 5.2.3

16. Final Architecture



17. Final Result

The complete web application environment was successfully configured on the RHEL server.

The setup includes:

- Apache HTTP Server
- PHP 8.3.33
- MariaDB
- sampledb database
- users table with sample records
- phpMyAdmin
- SELinux configuration for phpMyAdmin
- PHP information page

The PHP website was successfully accessed through:

<http://13.201.184.251/index.php>

The phpMyAdmin interface was successfully accessed through:

<http://13.201.184.251/phpmyadmin/>

The sampledb database and users table were successfully verified through phpMyAdmin.

18. Conclusion

Apache, PHP, and MariaDB were successfully installed and configured on the Red Hat Enterprise Linux server. A PHP information page was first hosted and verified through Apache, demonstrating successful PHP integration.

MariaDB was then configured with a dedicated database, user, table, and sample records. phpMyAdmin was subsequently installed to provide a web-based database management interface.

An SELinux context issue initially caused a 403 Forbidden response when accessing phpMyAdmin. The issue was resolved by assigning the appropriate `httpd_sys_content_t` context.

The final environment provides a functional **Apache + PHP + MariaDB + phpMyAdmin** stack accessible through the EC2 public IP **13.201.184.251**.